Predictive analytics using a Machine Learning Model to recommend the most suitable Intervention Technology for Autism related deficits.

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***Abstract*— This body of work aims to collect and analyse data from previous studies completed on technological interventions (to aid autism related deficits such as social behaviour, communication and limited interest and actions that are both distinct and repetitive) [1] for people with Autism and a build machine learning model for predicting the most suitable intervention technology for a single or combination of deficits related to Autism. The author selected and collected all relevant data from current available studies. This data was used to build and train supervised classification machine learning model to predict the most suitable intervention technology for a single or combination of deficits related to autism based on deficits presented by an individual.**

Results indicated that machine learning is an effective tool for building a predictive model to recommend the most effective intervention technology for Autism related deficits based on data integrated from the studies. The outcomes have implications for medical professionals, caregivers, teachers and family members in effectively selecting technological intervention for autism related deficits. These interventions could help the individual better cope with their disability and potentially lessen its impact.

Keywords—Data Analysis, Machine Learning, Supervised Classification, Ensemble Classifiers, Decision Tree, ASD interventions, Technological Interventions, Virtual Reality, Augmented Reality, Wearable assistive technology, Robot, Mobile, Serious games, Multi-media

I. INTRODUCTION

Autism is a very distressing and difficult problem not just for the sufferer but for their near and dear ones as well. At times it leaves an individual totally unaware of other people's state of mind and an individual can be either verbal or nonverbal. Autism covers a variety of conditions indicated in the form of deficits in social behaviour, communication and limited types of interest and actions which are distinct as well as repetitive [1]. It is prevalent among all ethnic, racial, and social groups at all stages of education, and is therefore indiscriminant. The occurrence of Autism is persistent around the world [2]. Autism is thus attracting a lot of attention and consideration in the field of research and development. Providentially, usage of information as well as communication technologies in teaching learners with learning difficulties like Autism have been shown to be very effective[3].

Consequently, technology has been put forth as an aid for teaching and learning in Autism. Computer-based interventions seem to have all it takes to be a potential tool in the years to come, given the economic costs, children's fascination with technology, and easy wide-ranging access. The natural attraction of kids with ASD towards computer technologies results in better participation and less disruption than in non-computer-based interventions [4]. Research to analyse current learning tools and systems could be of huge significance if some specific tools and systems are found to be more effective than others. In the same way, the lives of autistic people, their teachers and their families are also influenced by innovative support, improved learning prospects and better understanding of technological advancements and their effective application on an everyday basis [5].

This study, thus, aims at collecting and analysing relevant data to create a machine learning model intended for comparing and forecasting the 'most promising modes or technologies' as learning aids for individuals with ASD. Various data mining and analysing techniques using Python programming language are to be applied to get an insight into the data and find meaningful patterns, trends and more. Machine Learning algorithms will then be used to build a model for prediction and comparison.

II. MACHINE LEARNING AS AN AID FOR ASD

There have been many studies done to date on the use of machine learning as an aid for Autism. Some are discussed below:

A. Expediting Diagnosis and Screening of Autism

Wall et al. used machine learning to investigate if the screening and diagnosis time of Autism could be shortened. Autism Diagnostic Observation Schedule-Generic (ADOS) is among the most popular methods for the behavioural assessment of autism spectrum disorders. To date, ADOS is one of the foremost behavioural tools to verify a medical diagnosis. Attention is needed for diagnostic hold-ups and inconsistencies in the coverage of people due to the length of the ADOS examination and the requirement for supervision in a clinical facility by a trained professional. [6]

Huge assemblies of ADOS score sheets along with clinical diagnosis, important by-products of the extensive implementation and use of ADOS, amassed as a result of research efforts, can be used

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directly to address this objective [6]. Decision Tree and EnsembleVote Classifiers were built with machine learning, and specificity and sensitivity of each regarding diagnostic validity were objectively evaluated. Decision tree learning, which worked well to classify a wide range of individuals on and off the spectrum equally [6], was used. And consequential classifier was considerably briefer than the typical ADOS and identified quite a few behavioural forms that could lead to approaches to faster evaluation and diagnosis based on observation.

B. For classifying ASD and Typical development

Kakihara et al. used machine learning to investigate if the Classification of ASD and Typical Development could be made more accurate. Kakihara et al. reported the outcome of a classification experiment, on children with ASD utilising acoustic features, in which multiple kernel learning (MKL) algorithm is forwarded as a novel approach. Suggested MKL-SVM(Support Vector Machine) simultaneously evaluate both the weight of several acoustic features and classification borderline. The computed weight denotes how acoustic features are useful for the intended classification [7]. MKL-SVM's classification accuracy is 80.2% which is 9.5% higher than SVM 's accuracy, i.e. 70.7%. Furthermore, Kakihara et al.'s results show large weight acoustic characteristics largely in line spectral frequencies. In the classification of typically developing children and children with ASD, one of the formant features namely the line spectral frequency is also displayed as being weight[7].

C. For analysing movement

Perego et al. investigated the use of machine learning for movement analysis in autistic children to study neuro-motor functions in them. For this purpose, SVM is used to detect unusual aspects of reach and throw movements. Reach-and-throw movement has been chosen by Perego et al. as not only is it a landmark in the development of a child, but also its faulty or incomplete acquirement inhibit the development of every other superior neuro-motor functions. [8]

Perego et al. in the study examined the application of three different kernel functions of machine learning:

"• linear: $K(xi; xj) = xi \cdot xj$

• polynomial: $K(xi; xj) = (xi \cdot xj + 1)d$

• radial basis: $K(xi; xj) = \exp(-\sigma jxi - xjj2)$ " [8]

SVM classification proved its better performance (in comparison of classical statistical analysis). It also demonstrated that movement units are important features to be considered [8]. Considering the outcomes reach and throw movement can be used to detect early signs.

D. For recognising Autism Activity

Alwakeel et al. put forward that Autistic children can often develop unusual habits and sometimes, they could be unpredictable or rather dangerous to themselves and others. Due to inadequate communication and interaction, the unaware parents may underrate their bodily capabilities in comparison to their cognitive capabilities and may not understand that they could harm themselves. Therefore, there is great requirement for an automated alert system for parents/care takers assistance. So, an electronic system based on machine learning for Autism activity recognition with wireless sensor networks (WSNs) is presented by Alwakeel et al.

Autistic Child Sensor and Assistant System (ACSA), consist of three key parts: the companion ACSA Parent Application, the ACSA Wearable sensor device and the machine learning algorithms made for autistic movement event detection as well as processing [9]. The utility of machine learning into WSN is in "applying sensors environment data to improve the working of sensor networks in accomplishing various tasks without requiring software re-programming. Briefly, machine learning is significant for the integrations of multi WSNs, for example in impeded systems applications and in cyber physical systems." [9]

Detecting autistic child's motion and gestures accurately, is the main aim of the system. The device consists of several wearable WSN devices as well as the companion Parent Apps, along with associated machine learning algorithms aimed at Autism motion event detection. Capacity to detect a repetitive behaviour state in the autistic child is one feature of the device. This invention can thus be used by individuals and families having Autism Spectrum Disorder children as an alarming devices to help them to protect their autistic child [9]. The proposed system is anticipated to improve the life experience for all caretakers, the parents, the autistic child and the entire family.

E. For Personalizing Learning Material

Children having a neurological disorder like Autism require a personalised development system for their day-to-day activities. Designed for children having special needs, based on their varying traits, a research to deliver personalized learning materials is presented by Banik et al. "The system consisted of four parts: i) identifying level of the user by using machine learning algorithm. ii) web mining to generate multimodal learning materials from text story or learning keywords, iii) linking user preferences and Internet of things(IoT) enabled sensor data with the result, iv) personalising contents for users delineated with an intelligent interface." [10] The paper elucidates personalisation of outcomes using a machine learning algorithm.

F. For Predicting Mastery of Learning Outcomes

Linstead et al examined the use of machine learning to forecast the competence of learning outcomes. Thus, responding to behavioural therapy for Autistic individuals, Linstead et al implemented artificial neural networks to forecast learning outcomes the expertise of learning outcomes. Results for 726 children were reported. The linear regression models described in earlier studies being outperformed by neural networks and denotation of the advantages of applying yet sophisticated machine learning systems in the Autism research were presented in their results [11]. Linear regression methods to demonstrate the connection between therapy hours and learning results were the focus in previous studies, so to generate a baseline to collate in

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contrast to neural network performance Linstead et al. executed a similar regression using their log-transformed data. And using that they also presented the link of therapy periods with learning outcomes. Analysing further high-level machine learning approaches to forecast expertise of learning results became achievable with the established baseline. Intended for this a plain feed-forward neural network, was applied by Linstead et al. and backpropagation for 1000 epochs was used to train the networks. The largest analysis of the association between supervision hours, treatment intensity, gender and age to the expertise of learning results allotted in ABA regimens for children with ASD is represented in the study by Linstead et al.

The key inputs were application of neural networks that considerably enhanced on linear regression models, then improved interpretation of the importance of treatment strength concerning the learning process [11]. The reality that these approaches were being incorporated into a clinical setting too approves the significance of ML in the Autism research field. This study will take a different approach and investigate the possibility of using machine learning as a means of recommendation and selection of the intervention/technology that may help an individual with Autism most in their daily lives.

III. METHODOLOGY

This study covers the research on technological interventions to aid autism related deficits in an individual. Extensive research has been done into this subject matter and the development of a framework for recommending technological interventions. Currently, this area has had little focus. As a result of the lack of studies on comparison of technological interventions and published data. A considerable amount of time has been spent manually scraping studies to retrieve the required data. Data used in this study is the data of various studies on autism related interventions, the deficit they targeted, the system or technology they used and whether it gave 100% result or not. Some details about the study and data are given below:

Study Type: This study involves the statistical process of combining the findings from several individual studies. Thus, the type of this study is Meta-Analysis. Nearly 300 studies were reviewed. Meta-analysis of about 120 studies was attempted.

Study Area: This study covers the researches on technological interventions to aid autism related deficits in an individual. It takes into account, the technology used to aid a deficit and whether it gave cent percent result or not.

Target Population: The target population of this study is the collection of research on technological interventions developed to aid autism related deficits. The entire retrieved dataset (from several individual studies) of deficits and corresponding interventions for which the study data was used to make inferences is thus the target population of this study.

Study Population: The portion of population that the study accessed is a subset of the target population as it includes studies published from 2010 to 2018 only.

Sample Size estimation: As the samples are selected from a list of studies based on predefined setup and prerequisites it is probabilistic sampling.

Inclusion Criteria: Studies focusing on ASD/Autism related deficits, implementing technology based interventions and reporting clear outcomes (Where improvement in all the participants of the study is taken as 100% outcome and represented as 0, and less than 100% improvement i.e. that not all participants of the study improved is represented as 1).

Exclusion Criteria: studies excluded were: Non-English publication, publication year before 2010, not focusing on ASD/Autism, proposed works, inadequate in data, implementing non-technology based interventions and/or not having clear outcome.

Study Instrument: Selected parts of the reported results of chosen studies were entered into a database, and that "metadata'' was "meta-analysed", using meta-analysis as the study instrument, first descriptively and then inferentially to reach certain conclusions.

Data Collection Tool: A predefined CSV format with required variables was used for data collection. A template was devised to collect data, and from 120 research papers data was manually integrated into the template. In the binary columns of deficits and Improved, 0 represents 'Yes' and 1 represents 'No'.

The predefined CSV format had two types of variables, variables of study details and variables of analysis. Variables of study details give information of the studies included and the variables of analysis included data collected from the studies for analysis and prediction.

Variables of Study Details include-

- Paper Name, Author, Date, Publication, Intervention System, Variable measured, Aim and Link.

Variables of Analysis include-

- Serial number: A sequence of numbers starting from one. Serial number is assigned to identify individual study.
- Participants: Number of participants involved in the study.
- Communication and collaboration skills, Social Interaction, Eye contact, Emotion recognition/Emotional Intelligence, and Skill development (deficits): These variables denote the deficits under consideration. Each deficits variable is having either one or zero as a data point. After thoroughly reading each study, deficit columns were marked by either zero or one, where zero denotes positive and one denotes negative.
- Improved: After thoroughly reading each study, this column too is marked either zero or one. Where zero represents 100% outcome i.e. improvement in all the participants of the study. And one represents less than 100% improvement i.e. that not all participants of the study improved.
- Technology: This variable denotes the intervention technology used in each study.

Independent or predictor variables like autism related deficits and improvement criteria were used to measure dependent variable i.e. technology. This, study used predictor variable i.e. deficits to predict another variable or outcome, technology.

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Model Construction: In order to compare and predict the technology most suited for a deficit or a combination of deficits, two supervised machine learning models were developed, one based on the decision tree classification and the other on the Ensemble Vote Classification algorithm. [12]

From several supervised learning algorithms, Decision Tree Classifier was selected as the Decision Tree algorithm congregates various attributes by classifying them on the basis of their values and, classification is the foremost purpose for which it is used [13]. In comparison to individual learning models, a collection of learning models in the form of Ensemble Learning nearly always gives better performance. [14] Hence later EnsembleVoteClassifier was used to attain a better performance.

IV. Implementation

IEEE and Google Scholar databases were searched (up to September 2018) using a sequence of the following terms: Technological interventions, Virtual Reality, Augmented Reality, Wearable assistive technology, Robot, Mobile, Serious games, Multi-media, autism/ASD and/or PDD (pervasive developmental disorder). At first the headings and abstracts were read of all the papers from this initial search to finalise whether the paper warranted a more in-depth reading. If the study reported relevant outcome data on the technological intervention for autism(in terms of: target deficit(s) and improvement reported), the paper was obtained for more detailed study. Moreover, the reference sections of obtained papers were looked through to trace studies that might have been overlooked in the electronic search. The flow chart in figure-1 below illustrates the search and selection procedure.

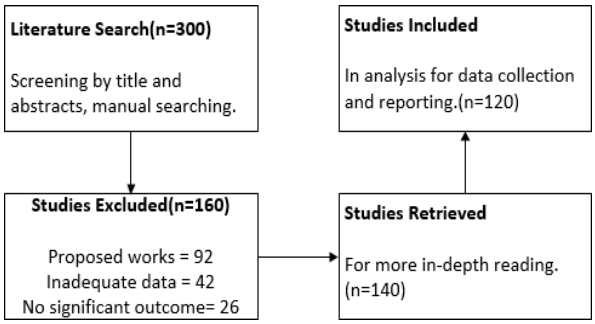


Figure 1 Steps of Data collection

The scope of this study was an extensive one. It not only involved selection of relevant studies to collect required data and then to feed the obtained data into a machine learning model to predict results. But implementation of relevant analysis techniques to obtain meaningful insights was also needed. The flow chart in figure-2, illustrates the design framework of the study.

- About 300 studies were found by means of keywords.
- 120 studies that were autism specific, deficit oriented, and had clear outcomes, these were selected and data was noted down into a pre-defined csv format. Zeros represent 'Yes', and Ones represent 'No' in the columns of deficits i.e. Communication skills, Emotion recognition and so on, and the column of Improved (100% or not).

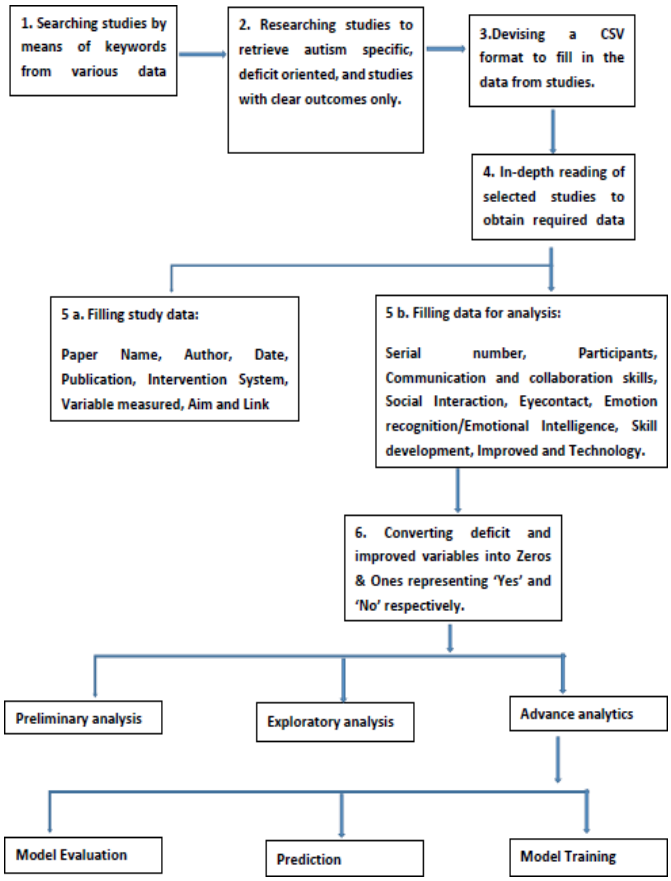


Figure 2 Study Design Framework

- Data was pre-processed to prepare it for model training, i.e., 'Variables of Study details' were dropped and only 'Variables of Analysis' were retained. Data was manually collected and hence did not require any cleaning processes. Preliminary analysis to study and analyse various variables was done with the use of Pandas Profiling function.
- Followed by Exploratory analysis, using various seaborn functions, to find several insights from the data like: Study-wise Technology Count, Technology-wise Patients Improvement results, Most Popular Technology for Each Deficit and Technology with maximum number of participants.
- Included in advance analysis were model development, prediction and evaluation. Data was then used to train the Decision Tree and EnsembleVoteClassifier. With the help of tree.DecisionTreeClassifier() and .fit function Decision Tree model was build and trained. In ML Ensemble techniques are the learning models which build a group of classifiers for the classification of new data points by means of obtaining a vote from their predictions. [15] This study uses following five models: DecisionTree, KNeighbors, LogisticRegression, RandomForest, GaussianNB and VotingClassifier is used to obtaining a (weighted) vote from their predictions.
- Using sklearn .values() and train_test_split() functions,.data slicing and model training was done. Lastly, with .predict() and model.score() function respectively, prediction and model

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evaluation was done.

V. RESULTS

Preliminary Analysis:

The objective of Preliminary data analysis was to reveal as much as possible about the structure i.e., key features, limitations and peculiarities etc., and to edit and prepare it for further analysis and modelling.

Results of Data Profiling: Dataset (after removing 'Variables of Study Details') has no missing values. In Means values of Participants Variable is 12.64. Technology Variable contain 30 robot-based, 34 Virtual/Augmented Reality, 17 Serious-game, 16 Mobile/Wearable assistive and 14 Multi-media tools technology. There are 8 variables and 120 observations. Dataset contains 6 Boolean, 1 Numeric, 1 categorical Variables. Count of 0(yes) and 1(No), for improvement in all participants of the study, in each Boolean variable is as shown below in Table I:

Communication and collaboration skills	0(yes)=37	1(No)=83
Emotion recognition/Intelligence	0(yes)=29	1(No)=91
Eye contact	0(yes)=30	1(No)= 90
Improved	0(yes)=103	1(No)= 17
Skill Development	0(yes)=33	1(No)= 87
Social interaction	0(yes)=70	1(No)= 50

Table I. Preliminary Findings

Exploratory Analysis:

Exploratory analysis is conducted to search for insights which might be beneficial in the context of prediction.

Study-wise Technology Count: The sns.countplot() fuction of seaborn gave count of each type of study. Showing that Virtual/ Augmented Reality and Robot based technology are the most studied technologies. Technology-wise Patients Improvement results: The pd.crosstab() function gave count of improvement outcomes for each technology. Maximum positive outcomes are given by Virtual/Augmented reality and Serious-game technology respectively. Most Popular Technology for Each Deficit: The .pivot_table() function was used to find most experimented or studied technology for each deficit. Robot based technology is most experimented or studied technology for Communication and collaboration skills, Emotion recognition/emotional intelligence, Skill development and Eyecontact. Whereas for social interaction Virtual/Augmented Reality is the most studied or experimented technology. Technology with maximum number of participants: With the help of .groupby() function count of participant each technology is applied on, was found. Virtual/Augmented Reality followed by Robot-based technology is most applied technology.

Advance Analysis:

A significant amount of research has been done on Autism and a lot of technological, non-technological and traditional interventions are studied. It was observed that numerous technologies are used to aid autism related deficits. Machine learning models were developed to predict technological intervention most suited for a single or a combination of deficits related to autism. The decision tree classifier, the first model, had 67% accuracy and the second model, the EnsembleVoteClassifier, had 75%.

The resultant models can predict most suitable technology for a single or a combination of deficits. Users can input deficit and Improvement Status combinations in the form of zeros and ones following the order of feature sequence (i.e., Communication and collaboration skills, Social interaction, Eye contact, Emotion recognition/Emotional Intelligence, Skill Development and Improvement Status) to obtain prediction of most suitable technology. Table II Below shows the sample of predictions given by the classifiers:

Deficits	**Decision tree classifier**	**EnsembleVote Classifier**
Communication	Virtual/Augmented reality	Serious game
Emotion recognition	Mobile/Wearable assistive technology	Virtual/Augmented Reality
Eye Contact	Serious game	Robot-based
Skill development	Serious game	Robot-based
Social interactions	Virtual/Augmented Reality	Virtual/Augmented Reality
(And for most) combinations of deficits	Robot-based technology	Robot-based technology

Table II. Predictions by the classifiers

From the evaluation of the findings, shown above in Table II, it can be concluded that for most combinations of deficits robot-based technology is most suited. Whereas most suited technologies for Communication, Emotion Recognition, Eye Contact, Skill development and Social Interaction are Serious Game, Virtual/Augmented Reality, Robot-based, Robot-based and Virtual/Augmented Reality respectively.

To conclude it can be said that with some improvements like inclusion of more studies and more features the EnsembleVoteClassifier model can be made more accurate and thus be effectively used to choose the best suited technology for single as well as combinations of deficits. Users may not have to try numerous technologies before they could find the most suited one. It can save a lot of time, money and energy of the users.

VI. CONCLUSION

In the presence of multitudes of technological interventions, it can be very tiresome and confusing to choose the most suitable and effective technology to aid an individual never mind an individual with complex deficits, such as a person with ASD. This machine learning model can help choose the most appropriate technology from the existing options based on studies and data integrated into the model.

This study involved the manual collection of all relevant data from various studies. No such study attempt had been made before and no precompiled data of such type was available. All data on technological interventions studied and used for deficits related to

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autism was collected. The data was then analysed to find patterns and details. Then the data was finally used to build the machine learning model.

The model developed is effectively recommending and predicting intervention technology for specific deficit or set of deficits based on learning acquired from the training data. Thus, it can have relevance in medical, caregiving, teaching and daily needs of autistic people. The results obtained from the study can be helpful in analysing various intervention technology in terms of outcomes, participants and deficits they targeted.

In order to complete this study nearly 300 studies were read and analysed to build content and extract data. Data from about 120 studies is integrated into the model to analyse various intervention technologies in terms of number of participants involved, deficits targeted, and outcome obtained.

Virtual reality, Augmented reality, Mobile-based technologies, Robot-based technologies, Serious games, Wearable assistive technologies were the major intervention technologies analysed in this study. Deficit wise analysis of intervention technologies ahead in terms of being more tried (no. of participants), more studied and in terms of results (100% improvement or not) was found.

Future work: For future work, the scope of this body of work could be expanded to include more studies, more variables or filters like age of the participants, time duration, and overall cost of the treatment/experiment could be considered. Studies on therapeutic interventions like Cognitive Behaviour Therapy, Lámh, PECS, and non-technological interventions could also be assessed to model their success rate on given individuals based on their presenting deficits.

Limitations: Readers should interpret the results with caution because of limited sample size (studies only from 2013 to 2018 are included) and factors like age, time duration and overall cost of interventions could not be included due to limited time and lack of data availability. Some of the major limitations were setting up, initiating and gathering manually all the data as no previous work or data focussed on this area of research previously. Finding related studies, then selecting relevant ones and extracting required data was time intensive and thus limited the ability to include some extra factors for the model to consider.

A major limiting factor was the lack of availability of studies with public results, more than 300 studies were researched but only 120 could be put to use for data extraction. Most of them were proposed studies or with inadequate data. Model's accuracy could have been improved if there were more studies with usable data.

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REFERENCES

[1] T. C. Theoharides and I. Tsilioni, 'Autism spectrum disorders', *Neuroimmune Pharmacology*, 2016. [Online]. Available: http://www.who.int/news-room/fact-sheets/detail/autism-spectrum-disorders. [Accessed: 09-Aug-2018].

[2] T. Rambhia, M. Dhodi, V. Patel, and D. R. Kalbande, 'Design of an intelligent system for autism', *Proc. - 2018 Int. Conf. Commun. Inf. Comput. Technol. ICCICT 2018*, vol. 2018-Janua, pp. 1–8, 2018.

[3] M. Shoaib, I. Hussain, H. T. Mirza, and M. Tayyab, 'The role of information and innovative technology for rehabilitation of children with Autism: A Systematic Literature Review', in *Proceedings of the 2017 17th International Conference on Computational Science and Its Applications, ICCSA 2017*, 2017, pp. 1–10.

[4] H. Zhao, A. R. Swanson, A. S. Weitlauf, Z. E. Warren, and N. Sarkar, 'Hand-in-Hand: A Communication-Enhancement Collaborative Virtual Reality System for Promoting Social Interaction in Children with Autism Spectrum Disorders', *IEEE Trans. Human-Machine Syst.*, vol. 48, no. 2, pp. 136–148, Apr. 2018.

[5] F. Shic and M. Goodwin, 'Introduction to Technologies in the Daily Lives of Individuals with Autism', *J. Autism Dev. Disord.*, vol. 45, no. 12, pp. 3773–3776, Dec. 2015.

[6] D. P. Wall, J. Kosmicki, T. F. Deluca, E. Harstad, and V. A. Fusaro, 'Use of machine learning to shorten observation-based screening and diagnosis of autism', *Transl. Psychiatry*, vol. 2, no. 4, pp. e100–e100, Apr. 2012.

[7] Y. Kakihara, T. Takiguchi, Y. Ariki, Y. Nakai, and S. Takada, 'Acoustic feature selection utilizing multiple kernel learning for classification of children with autism spectrum and typically developing children', in *2013 IEEE/SICE International Symposium on System Integration, SII 2013*, 2013, pp. 490–494.

[8] P. Perego, S. Forti, A. Crippa, A. Valli, and G. Reni, 'Reach and throw movement analysis with Support Vector Machines in early diagnosis of autism', in *Proceedings of the 31st Annual International Conference of the IEEE Engineering in Medicine and Biology Society: Engineering the Future of Biomedicine, EMBC 2009*, 2009, pp. 2555–2558.

[9] S. S. Alwakeel, B. Alhalabi, H. Aggoune, and M. Alwakeel, 'A machine learning based WSN system for autism activity recognition', in *Proceedings - 2015 IEEE 14th International Conference on Machine Learning and Applications, ICMLA 2015*, 2016, pp. 771–776.

[10] L. Banik, M. Bhuiyan, and A. Jahan, 'Personalized learning materials for children with special needs using machine learning', in *2015 Internet Technologies and Applications, ITA 2015 - Proceedings of the 6th International Conference*, 2015, pp. 169–174.

[11] E. Linstead, R. German, D. Dixon, D. Granpeesheh, M. Novack, and A. Powell, 'An application of neural networks to predicting mastery of learning outcomes in the treatment of autism spectrum disorder', in *Proceedings - 2015 IEEE 14th International Conference on Machine Learning and Applications, ICMLA 2015*, 2016, pp. 414–418.

[12] T. M. Mitchell, *Machine Learning*, vol. 61, no. 12. 1997.

[13] A. Dey, 'Machine Learning Algorithms: A Review', *Int. J. Comput. Sci. Inf. Technol.*, vol. 7, no. 3, pp. 1174–1179, 2016.

[14] D. Opitz and R. Maclin, 'Popular Ensemble Methods: An Empirical Study', *J. Artif. Intell. Res.*, vol. 11, pp. 169–198, Aug. 1999.

[15] T. G. Dietterich, 'Ensemble methods in machine learning', in *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 2000, vol. 1857 LNCS, pp. 1–15.

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